Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

If vectors $2\mathrm{i}-\mathrm{j}+\mathrm{k}, \mathrm{i}+2\mathrm{j}-3\mathrm{k}$ and $3\mathrm{i}+\mathrm{aj}+5\mathrm{k}$ are coplanar, then the value of a is

(A) 2

(B) -2

(C) -1

(D) -4

Q.2

If $\vec{\mathrm{m}}$, $\vec{\mathrm{n}}$ are non-parallel unit vectors and $\vec{\mathrm{r}}$ is a vector which is perpendicular to $\vec{\mathrm{m}}$ and $\vec{\mathrm{n}}$ such that $\left|\vec{\mathrm{r}}\right|=5$ and $\left|\vec{\mathrm{m}}+\vec{\mathrm{n}}\right|^2=2+4\left|\vec{\mathrm{m}}\times\vec{\mathrm{n}}\right|$, then the value of $\left|\left[\vec{\mathrm{m}}\ \vec{\mathrm{n}}\ \vec{\mathrm{r}}\right]\right|^2$ is equal to

(A) 7

(B) $\frac{21}{5}$

(C) 5

(D) $\frac{10}{3}$

Q.3

The position vector of a point in which a line through the origin perpendicular to the plane $2x-y-z=4$ meets the plane $\vec{r}\cdot(3\hat{i}-5\hat{j}+2\hat{k})=6$, is

(A) $(1,-1,-1)$

(B) $(-1,-1,2)$

(C) $(4,2,2)$

(D) $\left(\frac{4}{3},\frac{-2}{3},\frac{-2}{3}\right)$

Q.4

Let O be an interior point of triangle ABC, such that $2\overrightarrow{OA}+3\overrightarrow{OB}+4\overrightarrow{OC}=0$, then the ratio of the area of ΔABC to the area of ΔAOC is

(A) $3:1$

(B) $3:2$

(C) $2:1$

(D) $4:3$

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Q.5

The resultant of forces $\vec{P}$ and $\vec{Q}$ is $\vec{R}$. If $\vec{Q}$ is doubled, then $\vec{R}$ is doubled. If the direction of $\vec{Q}$ is reversed, then $\vec{R}$ is again doubled. Then $P^2 : Q^2 : R^2 =$

(A) $3 : 1 : 1$

(B) $2 : 3 : 2$

(C) $1 : 2 : 3$

(D) $2 : 3 : 1$

Q.6

Let $\vec{a}_1, \vec{a}_2 \ldots . \vec{a}_n$ be the sides of a regular polygon inscribed in a circle of unit radius. If $[\vec{a}_1 \times \vec{a}_2 + \vec{a}_2 \times \vec{a}_3 \ldots . . + \vec{a}_n \times \vec{a}_1] = [\vec{a}_1 \cdot \vec{a}_2 + \vec{a}_2 \cdot \vec{a}_3 + \ldots . . + \vec{a}_n \cdot \vec{a}_1]$ smallest possible value of n is

(A) 4

(B) 6

(C) 8

(D) 12

Q.7

Let $\mathrm{A}(-1, 0, 0), \mathrm{B}(0, 1, 0), \mathrm{C}(0, 0, -1)$ are the vertices of $\triangle \mathrm{ABC}$. If R and r denote the circumradius and inradius of $\triangle ABC$, then the value of $\left(R^2 + r^2\right)$ is equal to

(A) $\frac{5}{6}$

(B) $\frac{1}{6}$

(C) $\frac{2}{3}$

(D) $\frac{1}{3}$

Q.8

$ABCD$ is a parallelogram. Point P lies on side BC while point Q lies on CD such that $\overrightarrow{\mathrm{BP}} = 4\overrightarrow{\mathrm{PC}}$ and $\overrightarrow{\mathrm{CQ}} = 4\overrightarrow{\mathrm{QD}}$. The line PQ intersect the line AC at point E.
If $\overrightarrow{\mathrm{AE}} = \frac{\mathrm{k}_1}{\mathrm{k}_2}\overrightarrow{\mathrm{AC}}$, then value of $\mathrm{k}_1 + \mathrm{k}_2$ (where k_1 and k_2 are coprime numbers), is-

(A) 46

(B) 27

(C) 19

(D) 35

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Q.9

If $(\overline{\mathrm{R}}-\overline{\mathrm{S}})\cdot\hat{\mathrm{t}}=0,(\overline{\mathrm{R}}-\overline{\mathrm{S}})\cdot\hat{\mathrm{n}}=0$ and $(\overline{\mathrm{R}}-\overline{\mathrm{S}})\cdot\hat{\mathrm{b}}=0$, where $\hat{\mathrm{t}},\hat{\mathrm{n}},\hat{\mathrm{b}}$ form a right handed triad of orthogonal vectors, then -

(A) $\overline{\mathrm{R}}=\overline{\mathrm{S}}$

(B) $\overline{\mathrm{R}}\times\overline{\mathrm{S}}=\overline{0}$

(C) $\overline{\mathrm{R}}\cdot\overline{\mathrm{S}}=0$

(D) $\overline{\mathrm{R}}=0$

Q.10

Which of the following statement(s) is/are correct :

(A) If $\vec{a},\vec{b},\vec{c}$ are non-coplanar and $\vec{d}$ is any vector, then $[\vec{d}\vec{b}\vec{c}]\vec{a}+[\vec{d}\vec{c}\vec{a}]\vec{b}+[\vec{d}\vec{a}\vec{b}]\vec{c}-[\vec{a}\vec{b}\vec{c}]\vec{d}=\overrightarrow{0}$

(B) If I is incentre of $\triangle\mathrm{ABC}$ then $|\overrightarrow{\mathrm{BC}}|\overrightarrow{\mathrm{IA}}+|\overrightarrow{\mathrm{CA}}|\overrightarrow{\mathrm{IB}}+|\overrightarrow{\mathrm{AB}}|\overrightarrow{\mathrm{IC}}=\overrightarrow{0}$

(C) Any vector in three dimension can be written as linear combination of three non-coplanar vectors.

(D) In a triangle, if position vector of vertices are $\vec{a},\vec{b},\vec{c}$, then position vector of incentre is $\frac{\vec{a}+\vec{b}+\vec{c}}{3}$

Q.11

A line passes through the points whose position vectors are $\hat{i}+\hat{j}-2\hat{k}$ and $\hat{i}-3\hat{j}+\hat{k}$. The position vector of a point on it at a distance 2 units from the first point is -

(A) $\hat{i}-\frac{13}{5}\hat{j}+\frac{16}{5}\hat{k}$

(B) $\hat{i}+\frac{13}{5}\hat{j}-\frac{16}{5}\hat{k}$

(C) $\hat{i}+\frac{3}{5}\hat{j}+\frac{4}{5}\hat{k}$

(D) $\hat{\mathrm{i}}-\frac{3}{5}\hat{\mathrm{j}}-\frac{4}{5}\hat{\mathrm{k}}$

Q.12

Passage:
The position vectors of the vertices A,B and C of a tetrahedron ABCD are $(\hat{\mathrm{i}}+\hat{\mathrm{j}}+\hat{\mathrm{k}}),\hat{\mathrm{i}}$ and $3\hat{\mathrm{i}}$ respectively. The altitude from the vertex D to the opposite face ABC meets the median AF of $\triangle\mathrm{ABC}$ at $\mathrm{E}.\mathrm{AD}=4$ and $\mathrm{AE}=2\sqrt{3}$.
Question:
Area of $\Delta\mathrm{ABC}$

(A) $\sqrt{2}$ sq. units

(B) $2\sqrt{2}$ sq. units

(C) $\frac{1}{\sqrt{2}}$ sq. units

(D) $2\mathrm{sq}$. units

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Q.13

Passage:

The position vectors of the vertices A, B and C of a tetrahedron $ABCD$ are $(\hat{i}+\hat{j}+\hat{k}), \hat{i}$ and $3\hat{i}$ respectively. The altitude from the vertex D to the opposite face ABC meets the median AF of $\triangle ABC$ at $E. AD = 4$ and $AE = 2\sqrt{3}$.

Question:

Volume of the tetrahedron $DABC =$

(A) $\frac{\sqrt{2}}{3}$ cubic units

(B) $\frac{2\sqrt{2}}{3}$ cubic units

(C) $2\sqrt{2}$ cubic units

(D) $\frac{2}{3}$ cubic units

Q.14

Passage:

The position vectors of the vertices A, B and C of a tetrahedron $ABCD$ are $(\hat{i}+\hat{j}+\hat{k}), \hat{i}$ and $3\hat{i}$ respectively. The altitude from the vertex D to the opposite face ABC meets the median AF of $\triangle ABC$ at $E. AD = 4$ and $AE = 2\sqrt{3}$.

Question:

If F and E are on the same side of A, then the positive vector of E is -

(A) $3\hat{i}-\hat{j}-\hat{k}$

(B) $3\hat{i}+\hat{j}+\hat{k}$

(C) $-\hat{i}+3\hat{j}+3\hat{k}$

(D) $\hat{i}-3\hat{j}-3\hat{k}$

Q.15

Passage:

The three vectors $\vec{a}', \vec{b}', \vec{c}'$ defined by $\vec{a}' = \frac{\vec{b}\times\vec{c}}{[\vec{a}\,\vec{b}\,\vec{c}]}, \vec{b}' = \frac{\vec{c}\times\vec{a}}{[\vec{a}\,\vec{b}\,\vec{c}]}, \vec{c}' = \frac{\vec{a}\times\vec{b}}{[\vec{a}\,\vec{b}\,\vec{c}]}$, where $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar are called the reciprocal system to the vectors $\vec{a}, \vec{b}, \vec{c}$.

Question:

$\vec{a}\times a' + \vec{b}\times\vec{b}' + \vec{c}\times\vec{c}'$ equals $-$

(A) $\vec{0}$

(B) $\frac{\vec{a}+\vec{b}+\vec{c}}{[\vec{a}\,\vec{b}\,\vec{c}]}$

(C) $[\vec{a}\vec{b}\vec{c}](\vec{a}+\vec{b}+\vec{c})$

(D) None of these

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Q.16

Passage:

The three vectors $\vec{a}', \vec{b}', \vec{c}'$ defined by $\vec{a}' = \frac{\vec{b} \times \vec{c}}{[\vec{a}\,\vec{b}\,\vec{c}]}, \vec{b}' = \frac{\vec{c} \times \vec{a}}{[\vec{a}\,\vec{b}\,\vec{c}]}, \vec{c}' = \frac{\vec{a} \times \vec{b}}{[\vec{a}\,\vec{b}\,\vec{c}]}$, where $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar are called the reciprocal system to the vectors $\vec{a}, \vec{b}, \vec{c}$.

Question:

$\vec{a}' \times \vec{b}' + \vec{b}' \times \vec{c}' + \vec{c}' \times \vec{a}'$ equals –

(A) $\vec{a} + \vec{b} + \vec{c}$

(B) $\vec{a} \times (\vec{b} \times \vec{c})$

(C) $\frac{\vec{a}+\vec{b}+\vec{c}}{[\vec{a}\,\vec{b}\,\vec{c}]}$

(D) $(\vec{a} \times \vec{b}) \times \vec{c}$

Q.17

Passage:

The three vectors $\vec{a}', \vec{b}', \vec{c}'$ defined by $\vec{a}' = \frac{\vec{b} \times \vec{c}}{[\vec{a}\,\vec{b}\,\vec{c}]}, \vec{b}' = \frac{\vec{c} \times \vec{a}}{[\vec{a}\,\vec{b}\,\vec{c}]}, \vec{c}' = \frac{\vec{a} \times \vec{b}}{[\vec{a}\,\vec{b}\,\vec{c}]}$, where $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar are called the reciprocal system to the vectors $\vec{a}, \vec{b}, \vec{c}$.

Question:

$(\vec{a} + \vec{b} + \vec{c}) \cdot \left(\vec{a}' + \vec{b}' + \vec{c}'\right)$

(A) 3

(B) 2

(C) 1

(D) 0

Q.18

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$\vec{a}$ and $\vec{b}$ are two unit vectors inclined at an angle θ to each other. Then —

Column I	Column II
(A) $\lvert\vec{a}+\vec{b}\rvert<\sqrt{2}$	(p) $\frac{2\pi}{3}<\theta<\pi$
(B) $\lvert\vec{a}-\vec{b}\rvert<\sqrt{2}$	(q) $\frac{\pi}{3}<\theta<\frac{\pi}{2}$
(C) $\lvert\vec{a}+\vec{b}\rvert<1$	(r) $\frac{\pi}{2}<\theta<\pi$
(D) $1<\lvert\vec{a}-\vec{b}\rvert<\sqrt{2}$	(s) $0<\theta<\frac{\pi}{3}$
	(t) $0<\theta<\frac{\pi}{2}$

(A) (A) - s, (B) - q, (C) - r, (D) - t

(B) (A) - t, (B) - t, (C) - q, (D) - q

(C) (A) - r, (B) - p, (C) - p, (D) - r

(D) (A) - r, (B) - t, (C) - p, (D) - q

Q.19

Let O be an interior point of $\Delta\mathrm{ABC}$ such that $\overrightarrow{\mathrm{OA}}+2\overrightarrow{\mathrm{OB}}+3\overrightarrow{\mathrm{OC}}=\overrightarrow{0}$, then find the ratio of area of $\triangle\mathrm{ABC}$ to area of $\triangle\mathrm{AOC}$.

Q.20

A cube in the first octant has sides OP, OQ and OR of length 1, along the x-axis, y-axis and z-axis, respectively, where $O(0,\ 0,\ 0)$ is the origin. Let the centre of the cube be at $S\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$ and T be the vertex of the cube opposite to the origin O such that S lies on the diagonal OT. If $\overrightarrow{p}=\overrightarrow{SP}$, $\overrightarrow{q}=\overrightarrow{SQ}$, $\overrightarrow{r}=\overrightarrow{SR}$ and $\overrightarrow{t}=\overrightarrow{ST}$, then the value of $\left|\left(\overrightarrow{p}\times\overrightarrow{q}\right)\times\left(\overrightarrow{r}\times\overrightarrow{t}\right)\right|$ is :

Answers & Solutions

Q.1 Answer:

-4

Solution:

If given vectors are coplanar, then there exists two scalar quantities x and y such that $2\mathrm{i}-\mathrm{j}+\mathrm{k}=\mathrm{x}(\mathrm{i}+2\mathrm{j}-3\mathrm{k})+\mathrm{y}(3\mathrm{i}+\mathrm{aj}+5\mathrm{k})$ ....(1)

Comparing coefficient of i, j and k on both sides of (1) we get $x+3y=2, 2x+ay=-1, -3x+5y=1$ .....(2)

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Solving first and third equations, we get $x = 1/2, y = 1/2$

Since the vectors are coplanar, therefore these values of x **and** y **will satisfy the equation** $2x + ay = -1$

$\therefore \quad 2(1/2) + a(1/2) = -1 \Rightarrow a = -4$

Q.2 Answer:

5

Solution:

$\vec{r}$ **is parallel to** $\vec{m} \times \vec{n}$

Let θ **be angle between** $\vec{m}$ **and** $\vec{n}$

$\therefore \left|\vec{m} + \vec{n}\right|^2 = 2 + 4\left|\vec{m} \times \vec{n}\right|$

$1 + 1 + 2\cos\theta = 2 + 4 \times \sin\theta$

$\Rightarrow \tan\theta = \frac{1}{2}$

$\left|\left[\vec{m}\,\vec{n}\,\vec{r}\right]\right|^2 = \left|\vec{r}.\left(\vec{m} \times \vec{n}\right)\right|^2$

$= \left|\vec{r}\right|^2\left|\vec{m}\right|^2\left|\vec{n}\right|^2 \sin^2\theta$

$= 25 \times \frac{1}{5} = 5$

Q.3 Answer:

$\left(\frac{4}{3}, \frac{-2}{3}, \frac{-2}{3}\right)$

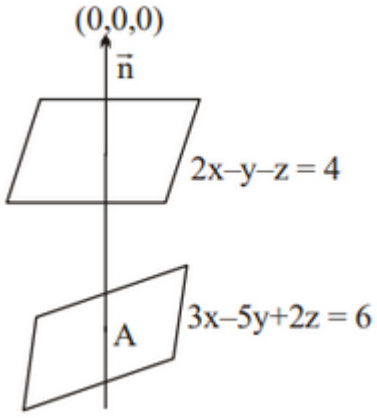
Solution:

Vector perpendicular to $2x - y - z = 4$ **is** $\vec{n} = 2\hat{i} - \hat{j} - \hat{k}$

Also line is collinear with $\vec{n}$**.**

So equation of line is $\vec{r} = \lambda(2\hat{i} - \hat{j} - \hat{k})$ **...(1)**

Now line (1) meet the plane $\vec{r} \cdot (3\hat{i} - 5\hat{j} + 2\hat{k}) = 6$



So, $\lambda(6 + 5 - 2) = 6$

$\Rightarrow \lambda = \frac{2}{3}$

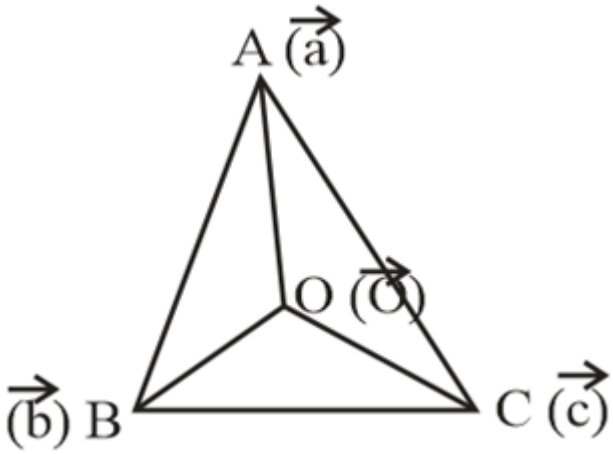
Hence p.v. of the point 'A' is $\frac{2}{3}(2\hat{i} - \hat{j} - \hat{k})$

Q.4 Answer:

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3 : 1

Solution:



Let position vectors of O, A, B, C are $\vec{O}, \vec{a}, \vec{b}, \vec{c}$ respectively.

$2\overrightarrow{OA} + 3\overrightarrow{OB} + 4\overrightarrow{OC} = 0$

$\Rightarrow 2\vec{a} + 3\vec{b} + 4\vec{c} = 0$

Area of $\Delta AOC = \frac{1}{2}\left|\overrightarrow{OA} \times \overrightarrow{OC}\right| = \frac{1}{2}\left|\vec{a} \times \vec{c}\right|$

Area of $\Delta ABC = \frac{1}{2}\left|\overrightarrow{BA} \times \overrightarrow{BC}\right| = \frac{1}{2}\left|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}\right|$

$= \frac{1}{2}\left|\vec{a} \times \left(\frac{-4\vec{c}-2\vec{a}}{3}\right) + \left(\frac{-4\vec{c}-2\vec{a}}{3}\right) \times \vec{c} + \vec{c} \times \vec{a}\right|$

$= \frac{1}{2}\left|\frac{4}{3}\vec{c} \times \vec{a} + \frac{2}{3}\vec{c} \times \vec{a} + \vec{c} \times \vec{a}\right|$

$= \frac{3}{2}\left|\vec{c} \times \vec{a}\right|$

$\frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta AOC} = \frac{3}{1}$

Q.5 Answer:

$2 : 3 : 2$

Solution:

$\vec{P} + \vec{Q} = \vec{R} \Rightarrow \overrightarrow{P^2} + \overrightarrow{Q^2} + 2\vec{P} \cdot \vec{Q} = \overrightarrow{R^2} \ldots (i)$

$|\vec{P} + 2\vec{Q}| = 2|\vec{R}| \Rightarrow \overrightarrow{P^2} + 4\overrightarrow{Q^2} + 4\vec{P} \cdot \vec{Q} = 4\overrightarrow{R^2} \ldots (ii)$

$|\vec{P} - \vec{Q}| = 2|\vec{R}| \Rightarrow \overrightarrow{P^2} + \overrightarrow{Q^2} - 2\vec{P} \cdot \vec{Q} = 4\overrightarrow{R^2} \ldots (iii)$

$(i) + (iii) \Rightarrow 2\overrightarrow{P^2} + 2\overrightarrow{Q^2} - 5\overrightarrow{R^2} = \overrightarrow{0} \ldots (iv)$

(ii) $+2($ **iii** $) \Rightarrow 3\vec{P}^2 + 6\vec{Q}^2 - 12\vec{R}^2 = \overrightarrow{0}$ **...(v)**

or $\vec{P}^2 + 2\vec{Q}^2 - 4\vec{R}^2 = \overrightarrow{0}$

By cross multiplication from (iv), (v), $\frac{\overrightarrow{p^2}}{2} = \frac{\overrightarrow{Q^2}}{3} = \frac{\overrightarrow{R^2}}{2}$.

Q.6 Answer:

8

Solution:

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$$|a_1 \times a_2| = |\vec{a}_1|\,|\vec{a}_2| \sin\frac{2\pi}{n}$$

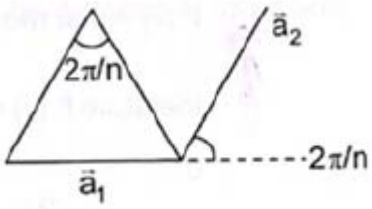
$$|\vec{a}_1 \cdot \vec{a}_2| = |\vec{a}_1|\,|\vec{a}_2| \cos\frac{2\pi}{n}$$

$$n\sin\frac{2\pi}{n} = n\cos\frac{2\pi}{n}$$

$$\tan\left(\frac{2\pi}{n}\right) = 1 \Rightarrow \frac{2\pi}{n} = m\pi + \frac{\pi}{4}$$

$$n = \frac{8}{1+4m}$$

Smallest possible value of n corresponds to $m = 0 \Rightarrow n = 8$



Q.7 Answer:

$\frac{5}{6}$

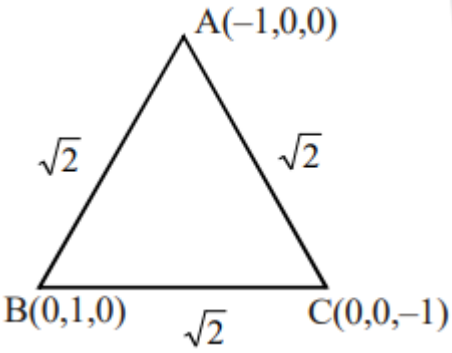
Solution:

Clearly $\triangle ABC$ is equilateral.

$$\Rightarrow \frac{r}{R} = 4 \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$$

$$\Rightarrow 2r = R$$

Also, $R = \frac{abc}{4\Delta} = \frac{2\sqrt{2}}{4 \cdot \frac{\sqrt{3}}{4} \cdot 2} = \sqrt{\frac{2}{3}} \Rightarrow r = \frac{1}{\sqrt{6}}$



$$\therefore \quad R^2 + r^2 = \frac{1}{6} + \frac{2}{3} = \frac{5}{6}$$

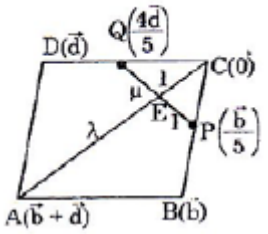
Q.8 Answer:

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Solution:

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$\overrightarrow{E} = \frac{\vec{b}+d}{\lambda+1} = \frac{\frac{\mu\vec{b}}{5}+\frac{4\vec{d}}{5}}{\mu+1}$
$\Rightarrow \mu = 4$ and $\lambda = \frac{21}{4}$
$\therefore \frac{\mu}{5(\mu+1)} = \frac{1}{\lambda+1}$ and $\frac{4}{5(\mu+1)} = \frac{1}{\lambda+1}$
$\overrightarrow{AE} = \left(\frac{\lambda}{\lambda+1}\overrightarrow{AC}\right) = \frac{21}{25}\overrightarrow{AC}$

Q.9 Answer:

$\overline{R} = \overline{S}$

Solution:

$(\overline{R} - \overline{S}) \cdot \hat{t} = 0$
$\Rightarrow$ **either** $\overline{R} - \overline{S} = 0$ **or** $\overline{R} - \overline{S}$ **is perpendicular to** $\hat{t}$.
Similarly, the other two conditions. But $\overline{R} - \overline{S}$ **cannot be perpendicular to all three** $\hat{t}, \hat{n}$ **and** $\hat{b}$ **since they are themselves mutually perpendicular.**
$\therefore \quad \overline{R} - \overline{S} = 0 \Rightarrow \overline{R} = \overline{S}$ and hence $\overline{R} \times \overline{S} = 0$

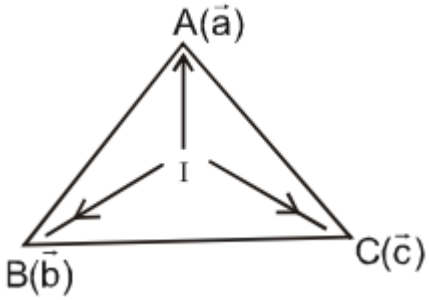
Q.10 Answer:

If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar and $\vec{d}$ is any vector, then $[\vec{d}\vec{b}\vec{c}]\vec{a} + [\vec{d}\vec{c}\vec{a}]\vec{b} + [\vec{d}\vec{a}\vec{b}]\vec{c} - [\vec{a}\vec{b}\vec{c}]\vec{d} = \overrightarrow{0}$

Solution:

(A) $[\vec{d}\vec{b} \quad \vec{c}]\vec{a} + [\vec{d}\vec{c}\vec{a}]\vec{b} + [\vec{d}\vec{a}\vec{b}]\vec{c} - [\vec{a}\vec{b}\vec{c}]\vec{d}$
$= ([\vec{b}\vec{c}\vec{d}]\vec{a} - [\vec{b}\vec{c}\vec{a}]\vec{d}) + ([\vec{d}\vec{a}\vec{b}]\vec{c} - [\vec{d}\vec{a}\vec{c}]\vec{b}) = (\vec{b} \times \vec{c}) \times (\vec{a} \times \vec{d}) + (\vec{d} \times \vec{a}) \times (\vec{c} \times \vec{b})$
$= (\vec{d} \times \vec{a}) \times (\vec{b} \times \vec{c}) - (\vec{d} \times \vec{a}) \times (\vec{b} \times \vec{c}) = \overrightarrow{0}$

(B)



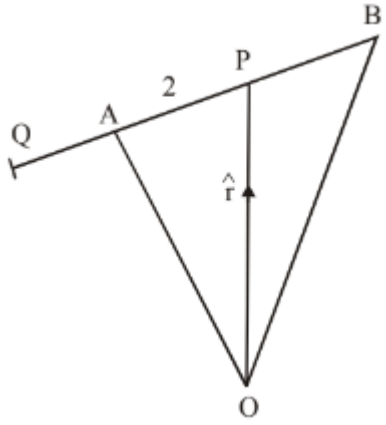
$I \equiv \frac{a\vec{a}+b\vec{b}+c\vec{c}}{a+b+c}$

Q.11 Answer:

$\hat{i} + \frac{13}{5}\hat{j} - \frac{16}{5}\hat{k}$

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Solution:



$$\overrightarrow{OP} = \vec{r} = \overrightarrow{OA} \pm 2 \cdot \frac{\overrightarrow{AB}}{|\overrightarrow{AB}|}$$

$$= \hat{i} + \hat{j} - 2\hat{k} \pm 2 \cdot \frac{\overrightarrow{OB} - \overrightarrow{OA}}{|\overrightarrow{OB} - \overrightarrow{OA}|} = \hat{i} + \hat{j} - 2\hat{k} \pm \frac{2(-4\hat{j} + 3\hat{k})}{|-4\hat{j} + 3\hat{k}|}$$

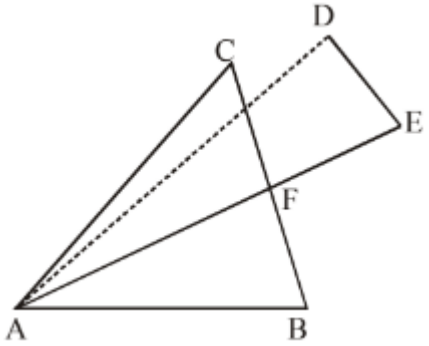
$$= \hat{i} + \hat{j} - 2\hat{k} \pm \frac{1}{5}(-8\hat{j} + 6\hat{k}) = \hat{i} + \left(1 - \frac{8}{5}\right)\hat{j} + \hat{k}\left(-2 + \frac{6}{5}\right)$$

or $\hat{i} + \left(1 + \frac{8}{5}\right)\hat{j} + \left(-2 - \frac{6}{5}\right)\hat{k}$ **i.e.,** $\hat{i} - \frac{3}{5}\hat{j} - \frac{4}{5}\hat{k}$ **or** $\hat{i} + \frac{13}{5}\hat{j} - \frac{16}{5}\hat{k}$

Q.12 Answer:

$\sqrt{2}$ sq. units

Solution:



(i) Area of $\Delta ABC = \frac{1}{2}|\overline{AB} \times \overline{AC}|$

$$= \frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -1 & -1 \\ 2 & -1 & -1 \end{vmatrix} = \frac{1}{2}|-2\hat{j} + 2\hat{k}| = \frac{1}{2} \times 2\sqrt{2} = \sqrt{2}$$ **sq. units**

Volume of the tetrahedron $= \frac{1}{3}$ **(area of** $\Delta ABC) \times$ **height** $= \frac{1}{3}(\sqrt{2}) \times 2 = \frac{2\sqrt{2}}{3}$ **cubic units**

(ii) $\quad AD = 4$ and $AE = 2\sqrt{3}$

Hence, $DE = \sqrt{16 - 12} = 2$

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(iii) P.V. of $F = 2\hat{i}$

$\overrightarrow{AF} = \hat{i} - \hat{j} - \hat{k}, \hat{AF} = \frac{\hat{i}-\hat{j}-\hat{k}}{\sqrt{3}}$

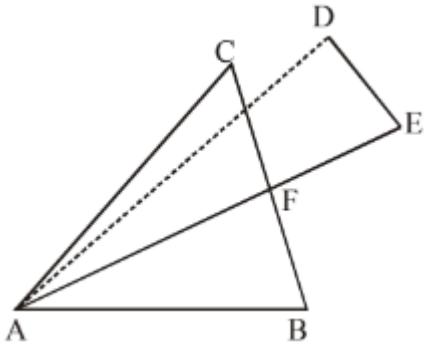
Since, $A = 2\sqrt{3}, \overrightarrow{AE} = 2\sqrt{3}\frac{(\hat{i}-\hat{j}-\hat{k})}{\sqrt{3}} = 2(\hat{i} - \hat{j} - \hat{k})$

P.V. of E = P.V. of $A + \overrightarrow{AE} = \hat{i} + \hat{j} + \hat{k} + 2(\hat{i} - \hat{j} - \hat{k}) = 3\hat{i} - \hat{j} - \hat{k}$

Q.13 Answer:

$\frac{2\sqrt{2}}{3}$ cubic units

Solution:



(i) Area of $\Delta ABC = \frac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}|$

$$= \frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -1 & -1 \\ 2 & -1 & -1 \end{vmatrix} = \frac{1}{2}|-2\hat{j} + 2\hat{k}| = \frac{1}{2} \times 2\sqrt{2} = \sqrt{2} \text{ sq. units}$$

(ii) $AD = 4$ and $AE = 2\sqrt{3}$

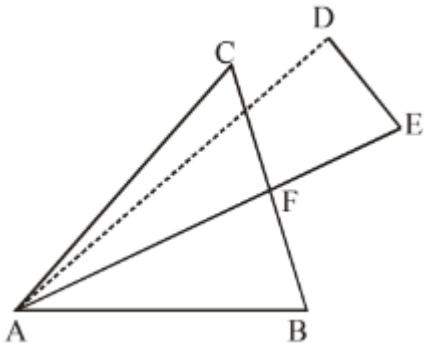
Hence, $DE = \sqrt{16 - 12} = 2$

Volume of the tetrahedron $= \frac{1}{3}($ **area of** $\Delta ABC) \times$ **height** $= \frac{1}{3}(\sqrt{2}) \times 2 = \frac{2\sqrt{2}}{3}$ **cubic units**

Q.14 Answer:

$3\hat{i} - \hat{j} - \hat{k}$

Solution:



(i) Area of $\Delta ABC = \frac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}|$

$$= \frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -1 & -1 \\ 2 & -1 & -1 \end{vmatrix} = \frac{1}{2}|-2\hat{j} + 2\hat{k}| = \frac{1}{2} \times 2\sqrt{2} = \sqrt{2} \text{ sq. units}$$

(ii)

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$\mathrm{AD}=4$ and $\mathrm{AE}=2\sqrt{3}$

Hence, $\mathrm{DE}=\sqrt{16-12}=2$

Volume of the tetrahedron $=\frac{1}{3}($ **area of** $\Delta \mathrm{ABC}) \times$ **height** $=\frac{1}{3}(\sqrt{2}) \times 2=\frac{2\sqrt{2}}{3}$ **cubic units**

(iii) P.V. of $\mathrm{F}=2\hat{\mathrm{i}}$

$\overline{\mathrm{AF}}=\hat{\mathrm{i}}-\hat{\mathrm{j}}-\hat{\mathrm{k}}, \hat{\mathrm{AF}}=\frac{\hat{\mathrm{i}}-\hat{\mathrm{j}}-\hat{\mathrm{k}}}{\sqrt{3}}$

Since, $\mathrm{A}=2\sqrt{3}, \overline{\mathrm{AE}}=2\sqrt{3}\frac{(\hat{\mathrm{i}}-\hat{\mathrm{j}}-\hat{\mathrm{k}})}{\sqrt{3}}=2(\hat{\mathrm{i}}-\hat{\mathrm{j}}-\hat{\mathrm{k}})$

$\text{P.V. of E} = \text{ P.V. of A} + \overline{\mathrm{AE}}=\hat{\mathrm{i}}+\hat{\mathrm{j}}+\hat{\mathrm{k}}+2(\hat{\mathrm{i}}-\hat{\mathrm{j}}-\hat{\mathrm{k}})=3\hat{\mathrm{i}}-\hat{\mathrm{j}}-\hat{\mathrm{k}}$

Q.15 Answer:

$\vec{0}$

Solution:

$\vec{\mathrm{a}} \times \vec{\mathrm{a}}'=\vec{\mathrm{a}} \times \frac{(\vec{\mathrm{b}} \times \vec{\mathrm{c}})}{[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]}=\frac{(\vec{\mathrm{a}} \cdot \vec{\mathrm{c}})\vec{\mathrm{b}}-(\vec{\mathrm{a}} \cdot \vec{\mathrm{b}})\vec{\mathrm{c}}}{[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]}$

Similarly, $\vec{\mathrm{b}} \times \vec{\mathrm{b}}'=\frac{(\vec{\mathrm{b}} \cdot \vec{\mathrm{a}})\vec{\mathrm{c}}-(\vec{\mathrm{b}} \cdot \vec{\mathrm{c}})\vec{\mathrm{a}}}{[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]}$ **and** $\vec{\mathrm{c}} \times \mathrm{c}'=\frac{(\vec{\mathrm{c}} \cdot \vec{\mathrm{b}})\vec{\mathrm{a}}-(\vec{\mathrm{c}} \cdot \vec{\mathrm{a}})\vec{\mathrm{b}}}{[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]}$

$\therefore \Sigma \vec{a} \times \vec{a}'=\vec{0} \quad (\because \vec{a} \cdot \vec{b}=\vec{b} \cdot \vec{a}$ **etc.** $)$

(ii) $\vec{\mathrm{a}}' \times \vec{\mathrm{b}}'=\frac{(\vec{\mathrm{b}} \times \vec{\mathrm{c}}) \times (\vec{\mathrm{c}} \times \vec{\mathrm{a}})}{[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]^2}=\frac{(\vec{\mathrm{b}} \times \vec{\mathrm{c}} \cdot \vec{\mathrm{a}})\vec{\mathrm{c}}-(\vec{\mathrm{b}} \times \vec{\mathrm{c}} \cdot \vec{\mathrm{c}})\vec{\mathrm{a}}}{[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]^2}=\frac{[\vec{\mathrm{b}}\,\vec{\mathrm{c}}\,\vec{\mathrm{a}}]\vec{\mathrm{c}}}{[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]^2}=\frac{\vec{\mathrm{c}}}{[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]}$

$(\because [\vec{\mathrm{b}}\,\vec{\mathrm{c}}\,\vec{\mathrm{a}}]=[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]$ **and** $\vec{\mathrm{b}} \times \vec{\mathrm{c}} \cdot \vec{\mathrm{c}}=0)$

Similarly, we can find $\vec{b}' \times \vec{c}'$ **and** $\vec{c}' \times \vec{a}'$ **as** $\frac{\vec{a}}{[\vec{a}\vec{b}\vec{c}]}$ **and** $\frac{\vec{b}}{[\vec{a}\vec{b}\vec{c}]}$ **respectively.**

$\therefore \sum \vec{\mathrm{a}}' \times \vec{\mathrm{b}}'=\frac{\vec{\mathrm{a}}+\vec{\mathrm{b}}+\vec{\mathrm{c}}}{[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]}$

(iii) $\vec{a} \cdot \vec{a}'=\frac{\vec{a} \cdot (\vec{b} \times \vec{c})}{[\vec{a}\vec{b}\vec{c}]}=\frac{[\vec{a}\vec{b}\vec{c}]}{[\vec{a}\vec{b}\vec{c}]}=1$ **and** $\vec{a} \cdot \vec{b}'=\frac{\vec{a} \cdot (\vec{c} \times \vec{a})}{[\vec{a}\vec{b}\vec{c}]}=0$

$\vec{\mathrm{a}} \cdot \vec{\mathrm{c}}'=0$ **in a similar way.**

Hence, $(\vec{a}+\vec{b}+\vec{c}) \cdot \left(\vec{a}'+\vec{b}'+\vec{c}'\right)=1+1+1=3.$

Q.16 Answer:

$\frac{\vec{\mathrm{a}}+\vec{\mathrm{b}}+\vec{\mathrm{c}}}{[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]}$

Solution:

$\vec{\mathrm{a}} \times \vec{\mathrm{a}}'=\vec{\mathrm{a}} \times \frac{(\vec{\mathrm{b}} \times \vec{\mathrm{c}})}{[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]}=\frac{(\vec{\mathrm{a}} \cdot \vec{\mathrm{c}})\vec{\mathrm{b}}-(\vec{\mathrm{a}} \cdot \vec{\mathrm{b}})\vec{\mathrm{c}}}{[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]}$

Similarly, $\vec{\mathrm{b}} \times \vec{\mathrm{b}}'=\frac{(\vec{\mathrm{b}} \cdot \vec{\mathrm{a}})\vec{\mathrm{c}}-(\vec{\mathrm{b}} \cdot \vec{\mathrm{c}})\vec{\mathrm{a}}}{[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]}$ **and** $\vec{\mathrm{c}} \times \mathrm{c}'=\frac{(\vec{\mathrm{c}} \cdot \vec{\mathrm{b}})\vec{\mathrm{a}}-(\vec{\mathrm{c}} \cdot \vec{\mathrm{a}})\vec{\mathrm{b}}}{[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]}$

$\therefore \Sigma \vec{a} \times \vec{a}'=\vec{0} \quad (\because \vec{a} \cdot \vec{b}=\vec{b} \cdot \vec{a}$ **etc.** $)$

(ii) $\vec{\mathrm{a}}' \times \vec{\mathrm{b}}'=\frac{(\vec{\mathrm{b}} \times \vec{\mathrm{c}}) \times (\vec{\mathrm{c}} \times \vec{\mathrm{a}})}{[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]^2}=\frac{(\vec{\mathrm{b}} \times \vec{\mathrm{c}} \cdot \vec{\mathrm{a}})\vec{\mathrm{c}}-(\vec{\mathrm{b}} \times \vec{\mathrm{c}} \cdot \vec{\mathrm{c}})\vec{\mathrm{a}}}{[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]^2}=\frac{[\vec{\mathrm{b}}\,\vec{\mathrm{c}}\,\vec{\mathrm{a}}]\vec{\mathrm{c}}}{[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]^2}=\frac{\vec{\mathrm{c}}}{[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]}$

$(\because [\vec{\mathrm{b}}\,\vec{\mathrm{c}}\,\vec{\mathrm{a}}]=[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]$ **and** $\vec{\mathrm{b}} \times \vec{\mathrm{c}} \cdot \vec{\mathrm{c}}=0)$

Similarly, we can find $\vec{b}' \times \vec{c}'$ **and** $\vec{c}' \times \vec{a}'$ **as** $\frac{\vec{a}}{[\vec{a}\vec{b}\vec{c}]}$ **and** $\frac{\vec{b}}{[\vec{a}\vec{b}\vec{c}]}$ **respectively.**

$\therefore \sum \vec{\mathrm{a}}' \times \vec{\mathrm{b}}'=\frac{\vec{\mathrm{a}}+\vec{\mathrm{b}}+\vec{\mathrm{c}}}{[\vec{\mathrm{a}}\,\vec{\mathrm{b}}\,\vec{\mathrm{c}}]}$

(iii) $\vec{a} \cdot \vec{a}'=\frac{\vec{a} \cdot (\vec{b} \times \vec{c})}{[\vec{a}\vec{b}\vec{c}]}=\frac{[\vec{a}\vec{b}\vec{c}]}{[\vec{a}\vec{b}\vec{c}]}=1$ **and** $\vec{a} \cdot \vec{b}'=\frac{\vec{a} \cdot (\vec{c} \times \vec{a})}{[\vec{a}\vec{b}\vec{c}]}=0$

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$\vec{a} \cdot \vec{c}' = 0$ **in a similar way.**

Hence, $(\vec{a}+\vec{b}+\vec{c}) \cdot \left(\vec{a}'+\vec{b}'+\vec{c}'\right) = 1+1+1 = 3.$

Q.17 Answer:

3

Solution:

$\vec{a} \times \vec{a}' = \vec{a} \times \frac{(\vec{b}\times\vec{c})}{[\vec{a}\,\vec{b}\,\vec{c}]} = \frac{(\vec{a}\cdot\vec{c})\vec{b}-(\vec{a}\cdot\vec{b})\vec{c}}{[\vec{a}\,\vec{b}\,\vec{c}]}$

Similarly, $\vec{b} \times \vec{b}' = \frac{(\vec{b}\cdot\vec{a})\vec{c}-(\vec{b}\cdot\vec{c})\vec{a}}{[\vec{a}\,\vec{b}\,\vec{c}]}$ **and** $\vec{c} \times c' = \frac{(\vec{c}\cdot\vec{b})\vec{a}-(\vec{c}\cdot\vec{a})\vec{b}}{[\vec{a}\,\vec{b}\,\vec{c}]}$

$\therefore \Sigma \vec{a} \times \vec{a}' = \vec{0} \quad (\because \vec{a}\cdot\vec{b} = \vec{b}\cdot\vec{a}$ **etc.** $)$

(ii) $\vec{a}' \times \vec{b}' = \frac{(\vec{b}\times\vec{c})\times(\vec{c}\times\vec{a})}{[\vec{a}\,\vec{b}\,\vec{c}]^2} = \frac{(\vec{b}\times\vec{c}\cdot\vec{a})\vec{c}-(\vec{b}\times\vec{c}\cdot\vec{c})\vec{a}}{[\vec{a}\,\vec{b}\,\vec{c}]^2} = \frac{[\vec{b}\,\vec{c}\,\vec{a}]\vec{c}}{[\vec{a}\,\vec{b}\,\vec{c}]^2} = \frac{\vec{c}}{[\vec{a}\,\vec{b}\,\vec{c}]}$

$(\because [\vec{b}\,\vec{c}\,\vec{a}] = [\vec{a}\,\vec{b}\,\vec{c}]$ **and** $\vec{b} \times \vec{c} \cdot \vec{c} = 0)$

Similarly, we can find $\vec{b}' \times \vec{c}'$ **and** $\vec{c}' \times \vec{a}'$ **as** $\frac{\vec{a}}{[\vec{a}\vec{b}\vec{c}]}$ **and** $\frac{\vec{b}}{[\vec{a}\vec{b}\vec{c}]}$ **respectively.**

$\therefore \sum \vec{a}' \times \vec{b}' = \frac{\vec{a}+\vec{b}+\vec{c}}{[\vec{a}\,\vec{b}\,\vec{c}]}$

(iii) $\vec{a} \cdot \vec{a}' = \frac{\vec{a}\cdot(\vec{b}\times\vec{c})}{[\vec{a}\vec{b}\vec{c}]} = \frac{[\vec{a}\vec{b}\vec{c}]}{[\vec{a}\vec{b}\vec{c}]} = 1$ **and** $\vec{a} \cdot \vec{b}' = \frac{\vec{a}\cdot(\vec{c}\times\vec{a})}{[\vec{a}\vec{b}\vec{c}]} = 0$

$\vec{a} \cdot \vec{c}' = 0$ **in a similar way.**

Hence, $(\vec{a}+\vec{b}+\vec{c}) \cdot \left(\vec{a}'+\vec{b}'+\vec{c}'\right) = 1+1+1 = 3.$

Q.18 Answer:

(A) - r, (B) - t, (C) - p, (D) - q

Solution:

$(A)-r,(B)-t,(C)-p,(D)-q$

(A) $|\vec{a}+\vec{b}|^2 < 2 \Rightarrow 1+1+2\vec{a}\cdot\vec{b} < 2 \Rightarrow \vec{a}\cdot\vec{b} < 0$

$\Rightarrow \cos\theta < 0 \Rightarrow \frac{\pi}{2} < \theta < \pi$

(B) $-2\cdot\vec{a}\cdot\vec{b} < 0 \Rightarrow \vec{a}\cdot\vec{b} > 0 \Rightarrow 0 < \theta < \frac{\pi}{2}$

(C)

$|\vec{a}+\vec{b}| < 1 \Rightarrow 2\vec{a}\cdot\vec{b} < -1 \Rightarrow \vec{a}\cdot\vec{b} < -\frac{1}{2} \Rightarrow \cos\theta < -\frac{1}{2}$

$\Rightarrow \quad -1 < \cos\theta < -\frac{1}{2} \Rightarrow \frac{2\pi}{3} < \theta < \pi$

(D) $1 < |\vec{a}-\vec{b}| < \sqrt{2} \Rightarrow |\vec{a}-\vec{b}| > 1$ **and** $|\vec{a}-\vec{b}| < \sqrt{2}$

$\Rightarrow \quad -2\vec{a}\cdot\vec{b} > -1$ **and** $-2\vec{a}\cdot\vec{b} < 0$

$\Rightarrow \quad \vec{a}\cdot\vec{b} < \frac{1}{2}$ **and** $\vec{a}\cdot\vec{b} > 0$

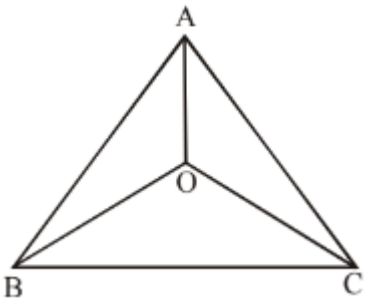
$\Rightarrow \quad \frac{\pi}{3} < \theta < \frac{\pi}{2}$

Q.19 Answer:

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Solution:

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$$\frac{\text{area of } (\Delta ABC)}{\text{area of } (\Delta OAC)} = \frac{\frac{1}{2}|\overrightarrow{OA}\times\overrightarrow{OB}|+|\overrightarrow{OB}\times\overrightarrow{OC}|+|\overrightarrow{OC}\times\overrightarrow{OA}|}{\frac{1}{2}|\overrightarrow{OA}\times\overrightarrow{OC}|} = \frac{|\overrightarrow{OA}\times\overrightarrow{OB}|+|\overrightarrow{OB}\times\overrightarrow{OC}|}{|\overrightarrow{OA}\times\overrightarrow{OC}|} + 1$$

$$\text{Given} : \overrightarrow{OA} + 2\overrightarrow{OB} + 3\overrightarrow{OC} = \overrightarrow{0}$$

$$2\overrightarrow{OB} \times \overrightarrow{OA} + 3\overrightarrow{OC} \times \overrightarrow{OA} = \overrightarrow{0}$$

$$\Rightarrow \frac{|\overrightarrow{OB} \times \overrightarrow{OA}|}{3} = \frac{|\overrightarrow{OC} \times \overrightarrow{OA}|}{2} = \lambda.....(1)$$

$$\overrightarrow{OB} \times \overrightarrow{OA} + 3\overrightarrow{OB} \times \overrightarrow{OC} = \overrightarrow{0}$$

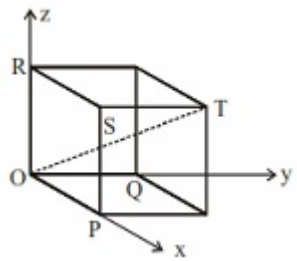
$$\Rightarrow \frac{|\overrightarrow{OB} \times \overrightarrow{OA}|}{3} = \frac{|\overrightarrow{OB} \times \overrightarrow{OC}|}{1} = \lambda.....(2)$$

$$\Rightarrow \text{Ratio of area of } \Delta \text{ s i.e., } \frac{\Delta ABC}{\Delta AOC} = \frac{3\lambda + \lambda}{2\lambda} + 1 = 3$$

Q.20 Answer:

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Solution:



$$\overrightarrow{p} = \overrightarrow{SP} = \left(\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}\right) = \frac{1}{2}\left(\hat{i} - \hat{j} - \hat{k}\right)$$

$$\overrightarrow{q} = \overrightarrow{SQ} = \left(-\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right) = \frac{1}{2}\left(-\hat{i} + \hat{j} - \hat{k}\right)$$

$$\overrightarrow{r} = \overrightarrow{SR} = \left(-\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}\right) = \frac{1}{2}\left(-\hat{i} - \hat{j} + \hat{k}\right)$$

$$\overrightarrow{t} = \overrightarrow{ST} = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right) = \frac{1}{2}\left(\hat{i} + \hat{j} + \hat{k}\right)$$

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$$\left|\left(\overrightarrow{p}\times\overrightarrow{q}\right)\times\left(\overrightarrow{r}\times\overrightarrow{t}\right)\right| = \frac{1}{4}\begin{vmatrix}\hat{i} & \hat{j} & \widehat{k}\\ 1 & -1 & -1\\ -1 & 1 & -1\end{vmatrix}\times\frac{1}{4}\begin{vmatrix}\hat{i} & \hat{j} & \widehat{k}\\ -1 & -1 & 1\\ 1 & 1 & 1\end{vmatrix}$$

$$= \frac{1}{16}\left|\left(2\hat{i}+2\hat{j}\right)\times\left(-2\hat{i}+2\hat{j}\right)\right| = \left|\frac{\widehat{k}}{2}\right| = \frac{1}{2}$$

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